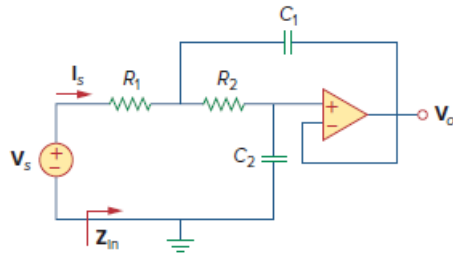


Problem

If the input impedance is defined as $Z_{in} = V_s/I_s$, find the input impedance of the op amp circuit in Fig. 10.116 when $R_1 = 10\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$, $C_1 = 10\text{ nF}$, $C_2 = 20\text{ nF}$, and $\omega = 5000\text{ rad/s}$.

Figure 10.116:



Step-by-step solution

Step 1 of 6

Refer to Figure 10.116 in the text book.

First convert the time domain circuit to frequency domain circuit.

$$\begin{aligned}
 X_{C_1} &= \frac{1}{j\omega C_1} \\
 &= \frac{1}{j \times 5000 \times 10 \times 10^{-9}} \\
 &= -j20\text{ k}\Omega \\
 X_{C_2} &= \frac{1}{j\omega C_2} \\
 &= \frac{1}{j \times 5000 \times 20 \times 10^{-9}} \\
 &= -j10\text{ k}\Omega
 \end{aligned}$$

Step 2 of 6

Draw the frequency domain circuit.

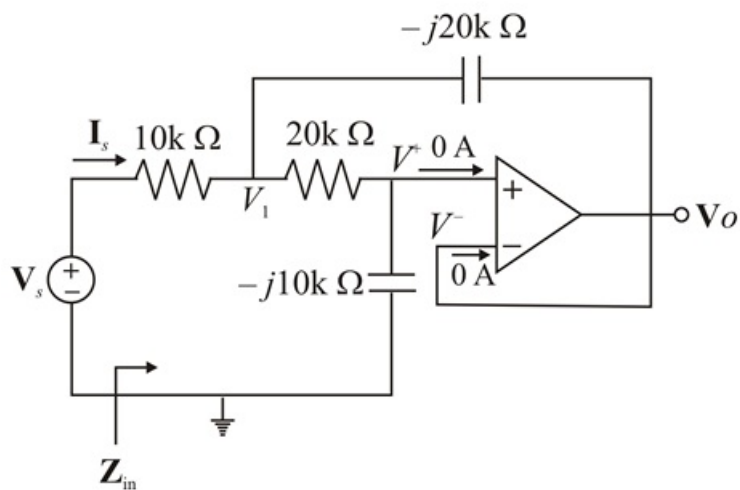


Figure 1

Step 3 of 6

From the above circuit,

$$V^- = V_o$$

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal, i.e.,

$$V^- = V^+$$

$$V^- = V^+ = V_o$$

Apply Kirchhoff's current law at node, V^+ .

$$\frac{V^+ - V_1}{20 \times 10^3} + \frac{V^+}{-j10 \times 10^3} = 0$$

$$\frac{V^+ - V_1}{20} + \frac{V^+}{-j10} = 0$$

$$\frac{V^+ - V_1}{20} + \frac{jV^+}{10} = 0 \quad \left(\text{Since, } \frac{1}{j} = -j \right)$$

$$\frac{V^+ - V_1 + j2V^+}{20} = 0$$

$$V^+ - V_1 + j2V^+ = 0$$

$$(1 + j2)V^+ - V_1 = 0$$

$$V_1 = (1 + j2)V^+$$

$$V_1 = (1 + j2)V_o \quad \left(\text{Since, } V^- = V^+ = V_o \right)$$

Thus, we get $V_1 = (1 + j2)V_o$

Step 4 of 6

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1 - V_s}{10 \times 10^3} + \frac{V_1 - V^+}{20 \times 10^3} + \frac{V_1 - V_o}{-j20 \times 10^3} = 0$$

$$\frac{V_1 - V_s}{10} + \frac{V_1 - V^+}{20} + \frac{j(V_1 - V_o)}{20} = 0$$

$$\frac{2(V_1 - V_s) + V_1 - V^+ + j(V_1 - V_o)}{20} = 0$$

$$2V_1 - 2V_s + V_1 - V^+ + jV_1 - jV_o = 0$$

$$(3 + j)V_1 - 2V_s - V^+ - jV_o = 0$$

Step 5 of 6

Substitute V_1 and V^+ in the equation.

$$(3 + j)((1 + j2)V_o) - 2V_s - V_o - jV_o = 0$$

$$(1 + j7)V_o - 2V_s - V_o - jV_o = 0$$

$$(1 + j7 - 1 - j)V_o - 2V_s = 0$$

$$(j6)V_o - 2V_s = 0$$

$$(j6)V_o = 2V_s$$

$$V_o = \frac{2}{j6}V_s$$

$$V_o = \left(\frac{1}{j3} \right) V_s$$

From Figure 1, write the expression for current, $\mathbf{I}_s$.

$$\mathbf{I}_s = \frac{\mathbf{V}_s - \mathbf{V}_1}{10 \times 10^3}$$

Substitute $\mathbf{V}_1$ in the equation.

$$\begin{aligned}\mathbf{I}_s &= \frac{\mathbf{V}_s - (1 + j2)\mathbf{V}_o}{10 \times 10^3} \\ &= \frac{\mathbf{V}_s - (1 + j2)\left(\left(\frac{1}{j3}\right)\mathbf{V}_s\right)}{10 \times 10^3} \\ &= \frac{(j3 - 1 - j2)\mathbf{V}_s}{j3(10 \times 10^3)} \\ &= \frac{(-1 + j1)\mathbf{V}_s}{j3(10 \times 10^3)}\end{aligned}$$

$$\frac{\mathbf{V}_s}{\mathbf{I}_s} = \frac{j3(10 \times 10^3)}{-1 + j1}$$

$$\begin{aligned}\mathbf{Z}_{in} &= \frac{j30 \times 10^3}{-1 + j1} \\ &= \frac{30 \times 10^3 \angle 90^\circ}{1.414 \angle 135^\circ} \\ &= 21.21 \times 10^3 \angle -45^\circ \\ &= 21.21 \angle -45^\circ \text{ k}\Omega\end{aligned}$$

Therefore, the input impedance, $\mathbf{Z}_{in}$ is $21.21 \angle -45^\circ \text{ k}\Omega$.